Comparative Evaluation of Statistical, Machine Learning, and Deep Learning Models for Daily Solar Radiation Forecasting in Dublin, Ireland

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Abstract

Accurate solar radiation forecasting is important for understanding solar-resource variability and supporting renewable-energy applications. This study presents a comparative evaluation of statistical, machine learning, and deep learning models for daily solar radiation forecasting using ten years of historical meteorological data from Dublin, Ireland, covering 2015–2024. Three forecasting models, Prophet, XGBoost, and Long Short-Term Memory (LSTM), are evaluated using meteorological predictors and engineered temporal features. The analysis focuses on the model's performance assessed by coefficient of determination (R^2), mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). The results show that XGBoost achieved the strongest performance with $R^2 = 0.8524$, MAE = 21.1591, and RMSE = 28.6734. Prophet was competitive, obtaining $R^2 = 0.8389$, MAE = 22.7991, and RMSE = 29.9600, while LSTM obtained $R^2 = 0.4679$, MAE = 36.1265, and RMSE = 54.4403. The findings indicate that greater model complexity does not necessarily result in improved accuracy.

Keywords: solar radiation forecasting; renewable energy; time-series forecasting; XGBoost; Prophet; LSTM; machine learning; deep learning; Dublin

1 Introduction

Solar radiation is a fundamental variable for solar-energy applications because it characterises the availability of solar resources. Forecasting solar radiation is challenging because atmospheric conditions produce nonlinear and time-dependent variation, while seasonal patterns influence the magnitude, and temporal structure of the resource.

Machine learning, and deep learning methods have increasingly been applied to environmental and renewable-energy forecasting. However, increasing the complexity of the model does not necessarily guarantee better predictive performance, particularly when datasets are moderate in size and contain a strong seasonal structure.

This study evaluates three forecasting paradigms (Prophet, XGBoost, and Long Short-Term Memory (LSTM)) for daily solar radiation forecasting in Dublin, Ireland. The research uses ten years of historical meteorological observations from 2015 to 2024 and compares the models under a common cleaned-data experimental setting.

The research question is: *How do statistical, machine learning, and deep learning models (Prophet, XGBoost, and LSTM) compare in terms of forecasting accuracy when applied to historical weather data?*

2 Related Work

2.1 Solar Radiation Forecasting

Solar radiation forecasting is relevant to renewable-energy planning and operation because temporal knowledge of solar-resource availability can support energy management and solar-energy applications. Forecasting performance is influenced by atmospheric variability, seasonal behaviour, feature availability, and forecast horizon.

2.2 Statistical, Machine Learning, and Deep Learning Approaches

Prophet provides a decomposable forecasting framework that models trend, and seasonal components and can incorporate additional regressors [Taylor and Letham, 2018]. XGBoost is a gradient-boosted decision-tree framework designed to efficiently model nonlinear relationships and interactions between predictors [Chen and Guestrin, 2016]. LSTM networks are recurrent neural architectures designed to learn temporal dependencies through gated memory mechanisms [Hochreiter and Schmidhuber, 1997].

3 Dataset and Study Area

The dataset consists of daily historical weather observations for Dublin, Ireland. Data were obtained through the Visual Crossing Weather API using the coordinates 53.3481 latitude and -6.24827 longitude. The observation period extends from 1 January 2015 to 31 December 2024.

The available variables include temperature-related measurements, humidity, precipitation, wind variables, atmospheric measurements, cloud cover, visibility, and solar-related variables. Solar radiation was selected as the target variable.

The study area provides a temperate maritime setting characterised by substantial cloud cover, seasonal variation, and day-to-day meteorological variability.

4 Data Preparation

Data preparation included quality assessment, missing-value handling, outlier screening, exploratory analysis, and feature selection.

The analysis identified strong annual seasonality in solar radiation as Figure 1 illustrates. Meteorological variables such as temperature, humidity, cloud cover, and wind-related measurements were considered as potential predictors.

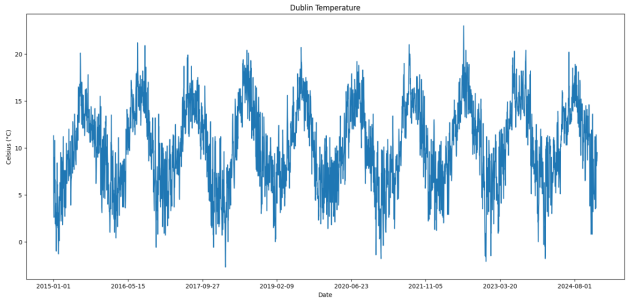


Figure 1: showing a Dublin's temperature time series. Strong seasonality is observed.

Variables with strong correlation or potential information leakage were excluded where appropriate, as observed in Figure 2. Solar energy and UV index were not used as predictors because of their strong relationship with solar radiation.

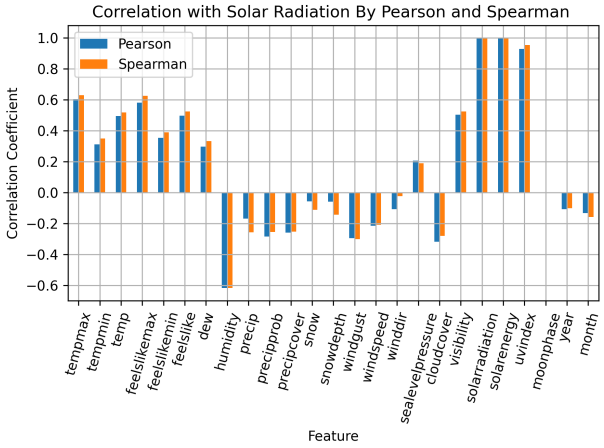


Figure 2: correlation between features and solar radiation. Strong relationship is observed across 'solar energy' and 'UV index'

Physically implausible observations were examined using statistical screening and domain validation, observe Figure 3. Genuine weather extremes were retained, while implausible observations were corrected where justified.

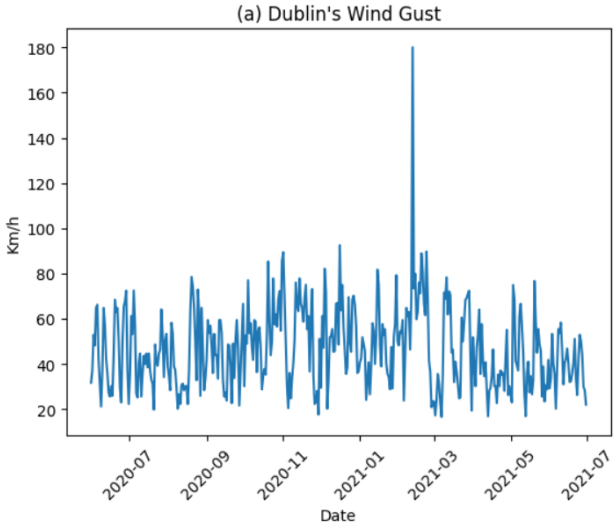


Figure 3: implausible observation in Dublin's wind gust feature.

The data were divided chronologically, with the oldest 80% used for model training and the most recent 20% used for testing, as illustrated in Figure 4.



Figure 4: split on the data. The fist chronologically 80% is intended for training the models, while the 20% resting data will be used for testing purposes.

5 Forecasting Models

5.1 Prophet

Prophet was used as the representative statistical forecasting model. The selected primary configuration used linear growth, additive seasonality, yearly seasonality of 16 Fourier terms, weekly seasonality of 5 Fourier terms, and tuned changepoint and prior-scale parameters.

5.2 XGBoost

XGBoost was used as the representative machine learning model. The model incorporated temporal and meteorological predictors. Hyperparameter optimisation was performed using randomized search. The selected primary configuration included 1317 estimators, maximum depth of 3, learning rate of approximately 0.0242, and tuned subsampling and column-sampling parameters.

5.3 Long Short-Term Memory

LSTM was used as the deep learning model. The network learned sequential relationships from scaled input sequences. The selected primary configuration used 96 recurrent units, a learning rate of 0.003, and a dense layer containing 32 units. Early stopping was used during training.

5.4 Evaluation Metrics

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{1}$$

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{2}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}. \tag{3}$$

$$MAPE = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \tag{4}$$

6 Results

Table 1 presents model's performance on daily solar radiation forecasting.

Table 1: Performance of Prophet, XGBoost, and LSTM.

Model	R^2	MAE	RMSE	MAPE
XGBoost	0.8524	21.1591	28.6734	0.3577
Prophet	0.8389	22.7991	29.9600	0.4812
LSTM	0.4679	36.1265	54.4403	0.5604

XGBoost achieved the highest R^2 and the lowest MAE and RMSE among the three models. Prophet produced results close to XGBoost, while LSTM showed substantially lower explanatory performance and higher absolute errors.

6.1 Visualisation Results

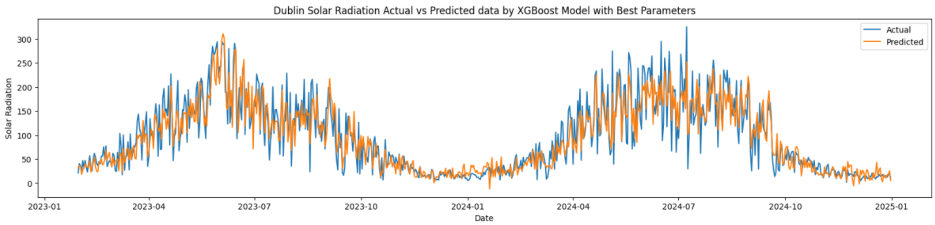


Figure 5: Forecast vs Actual data by XGBoost Model.

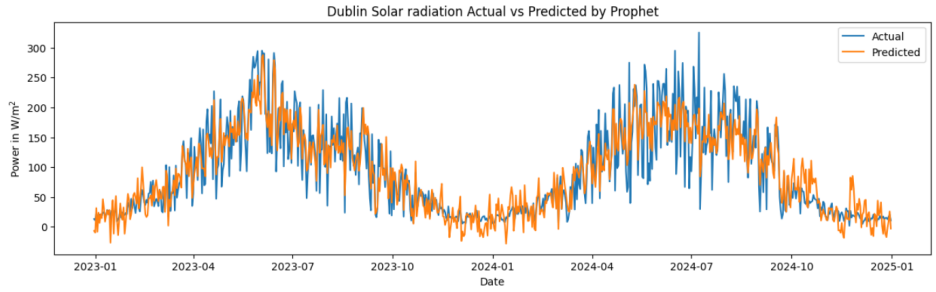


Figure 6: Forecast vs Actual data by Prophet Model.

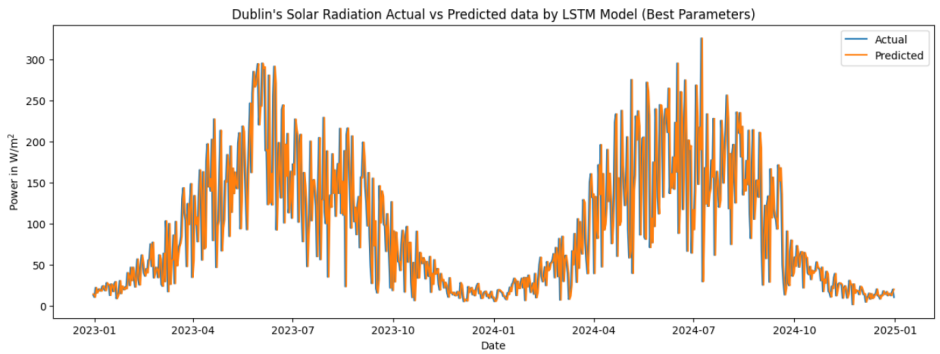


Figure 7: Forecast vs Actual data by LSTM Model.

7 Discussion

This experiment indicates that XGBoost was the strongest-performing approach for the Dublin solar-radiation dataset. Its performance is consistent with the ability of tree-based ensemble methods to represent nonlinear relationships between meteorological predictors and the target variable.

Prophet also performed competitively. Its explicit representation of seasonal structure is relevant to solar radiation because the target exhibits strong annual patterns.

LSTM performed less effectively than both XGBoost and Prophet. The underlying research suggests that the moderate dataset size, daily variability, and noise may limit the ability of the deep learning architecture to learn useful temporal dependencies.

The results therefore suggest that model complexity alone is not sufficient to guarantee forecasting improvement. For this dataset, a carefully tuned machine learning model provided stronger predictive performance than the evaluated deep learning model.

8 Implications for Solar-Energy Forecasting

Solar-radiation forecasting can provide information about temporal solar-resource availability and can therefore support downstream renewable-energy applications. The results suggest that relatively efficient machine learning approaches can be competitive with more complex deep learning architectures when historical data are moderate in size and exhibit strong seasonal structure.

The findings are specific to the Dublin dataset and should not be interpreted as evidence that XGBoost will outperform LSTM in all solar-energy forecasting settings.

9 Limitations

The study uses a single geographic location, Dublin, Ireland, and therefore does not establish geographical generalisation across Ireland or other climates. The dataset is based on daily observations, so the findings do not directly address intraday or hourly forecasting.

The comparison is limited to three representative modelling approaches. Additional algorithms and forecasting baselines could provide a broader benchmark.

Because solar radiation can contain values close to zero, percentage-based metrics such as MAPE require cautious interpretation.

10 Conclusion

This study compared Prophet, XGBoost, and LSTM for daily solar radiation forecasting using ten years of meteorological data from Dublin, Ireland. XGBoost achieved the strongest performance, Prophet produced competitive results, and LSTM performed less effectively.

The findings demonstrate that a more complex deep learning architecture does not necessarily provide superior forecasting accuracy for a moderate-sized, strongly seasonal meteorological dataset. The results support systematic model comparison and careful feature engineering when selecting forecasting approaches for solar-resource applications.

Data Availability

The study used historical meteorological data obtained through the Visual Crossing Weather API. Availability and reuse of the underlying data should be verified against the provider's current terms

before publication.

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